

Project Architecture and Development Guide

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This page explains build and deployment, package managers, CI, Docker images, and service wiring for the Robot Visual Perception project.

Overview

The system consists of three main components:

- **Frontend** (Vite + React)
- **Streamer service** (camera capture + WebRTC streaming)
- **Analyzer service** (inference, metadata generation)

The frontend connects **only to the Analyzer**.

The Analyzer establishes its own WebRTC connection to the Streamer.

Package Managers

Frontend

- **Node 20 + npm**
- **Commands:**
 - `npm install / npm ci`
 - `npm run build`
- **Used for:** Vite + React tooling

Backend

- **Python 3.11 with uv**
 - `uv python install 3.11`
 - `uv sync --extra dev --extra inference --extra onnx-tools --extra onnx-cpu`
- **Used for:** fast Python env creation and reproducible builds (same as CI)

Makefile Overview

The Makefile provides a single unified interface for all development tasks. CI workflows also call these same targets.

Common Targets

- **Install**
 - `make install`
 - `make install-frontend`
 - `make install-backend`
- **Lint**

- `make lint`
- `make lint-frontend`
- `make lint-backend`
- `make lint-licensing`
- `make format`
- `make format-check`

- **Test**

- `make test`
- `make test-frontend`
- `make test-backend`

- **Local Runs**

- `make run-streamer-local`
- `make run-analyzer-local`
- `make run-frontend-local`

- **Docker**

- `make docker-build[-frontend|-backend]`
- `make docker-compose-up`
- `make docker-compose-down`

Run `make help` to see all targets.

Continuous Integration (GitHub Actions)

File: `.github/workflows/ci.yml` Triggers: push and pull requests on `main`.

Pipeline Stages

Lint

- REUSE license compliance
- Frontend: `npm run lint`
- Backend (uv): `ruff + mypy`

Build

- Frontend: `npm run build`
- Backend: import check

```
uv run python -c "import streamer.main"
uv run python -c "import analyzer.main"
```



Test

- Frontend: `make test-frontend`
- Backend: `make test-backend`

Package

- Docker images built via:
 - `make docker-build-frontend`
 - `make docker-build-backend`

Docker Compose Test

- Builds all images
- Boots the full stack
- Hits `/health` endpoints

- Verifies Analyzer → Streamer connectivity

Docker Images

Frontend

- Build: `src/frontend/Dockerfile`
- Image: `robot-frontend:latest`
- Serves static build on **port 80**

Streamer Service

- Dockerfile: `Dockerfile.streamer`
- Image: `robot-streamer:latest`
- FastAPI on **8000**

Analyzer Service

- Dockerfile: `Dockerfile.analyzer`
- Image: `robot-analyzer:latest`
- FastAPI on **8001**
- Connects to the Streamer via `STREAMER_OFFER_URL`

Service Architecture

Streamer Service (8000)

- Accepts WebRTC offers from the frontend
- Streams raw camera frames

Analyzer Service (8001)

- Accepts WebRTC offers from the frontend
- Creates its own WebRTC session to webcam:
 - `STREAMER_OFFER_URL=http://localhost:8000/offer`
- Runs detection + overlay
- Has metadata data channel (placeholder)

Frontend

- Uses `VITE_BACKEND_URL` → `analyzer /offer`

Local Development

Install

```
make install
```



Lint

```
make lint
```



Test

```
make test
```



Run Services (3 terminals)

Streamer

```
make run-streamer-webcam
```



Analyzer

```
make run-analyzer-local
```



Frontend

```
make run-frontend-local
```



Deployment Notes

Hardware

- For Torch or ONNX GPU inference, install matching CUDA or ROCm drivers

Environment Variables

- VITE_BACKEND_URL
- STREAMER_OFFER_URL
- DETECTOR_BACKEND
- TORCH_DEVICE
- ONNX_PROVIDERS

Networking

- Expose/route:
 - Frontend: 8080 → 80
 - Webcam: 8000
 - Analyzer: 8001

HTTPS

Terminate TLS at the proxy/ingress; backend services run plain HTTP.

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